



EVOLUTION 2026 · MACROEVOLUTION

The tempo & mode of body size evolution in charontid whip spiders

Disentangling climatic and habitat drivers of an ancient, conserved body plan

Paulo M. Martins · Gabriela A. Galvão · **Diogo B. Provete** · Thiago Gonçalves-Souza · Gustavo S. de Miranda

The evolutionary paradox

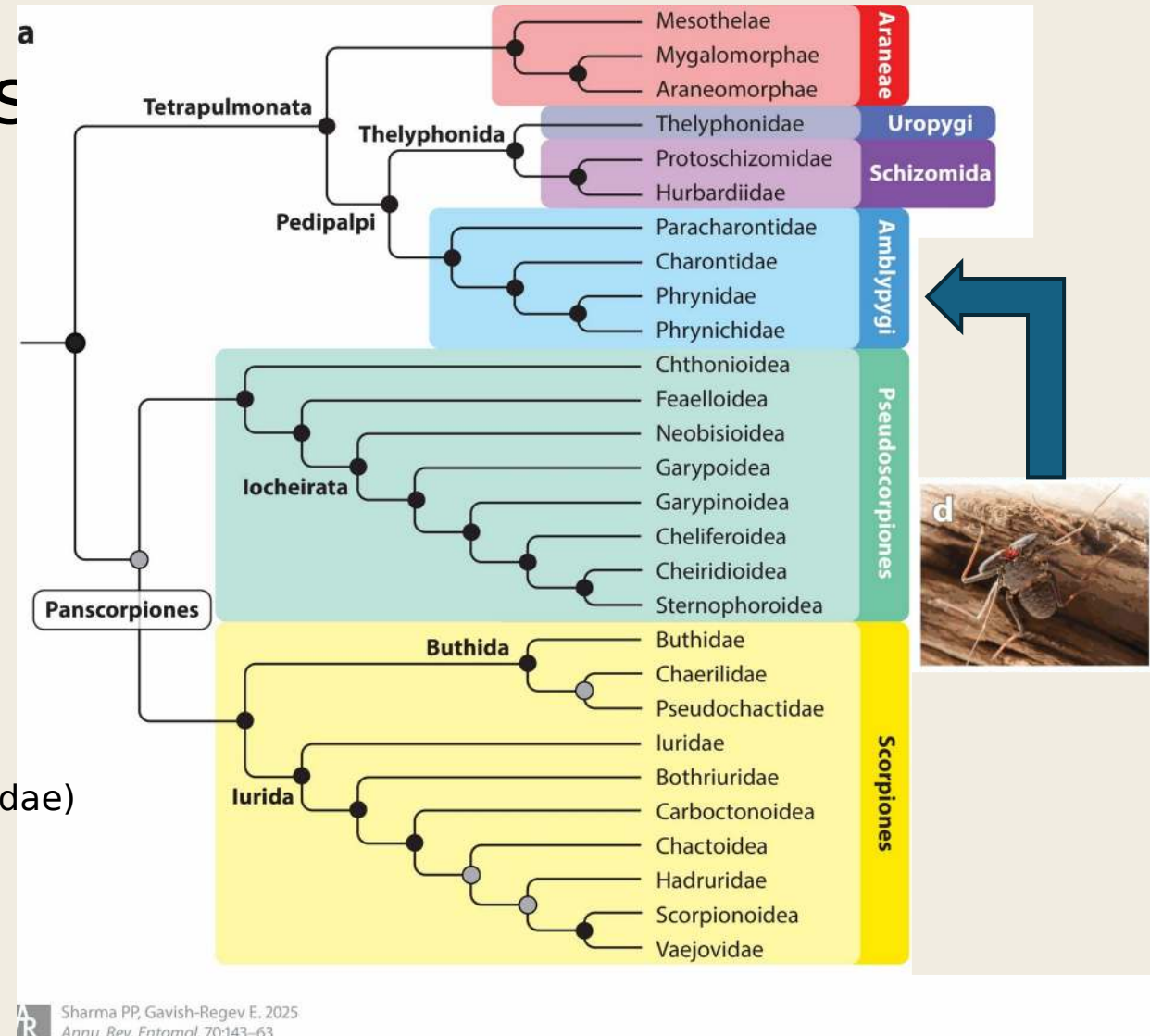
Amblypygi (Whip Spiders) is an arachnid order that emerged over **330 million**



Charinus magalhaesi (Charontidae)
SE Brazil
Photo: Gustavo S de Miranda



Heterophrynus longicornis (Phrynididae)
Central Amazon, Brazil



Weygoldtina anglica
(Oldest known fossil)
Carboniferous
Haug & Haug 2021



Heterophrynus longicorn
(modern, Phrynidae)
Amazon, Brazil

Corporate needs you to find the differences
between this picture and this picture.



They're the same picture.

Tackling the evolutionary paradox

What drives this apparent stasis?

Why such extreme morphological conservatism?

Is it stabilizing selection or a "jack-of-all-trades" strategy?

Why body size?

Body size is a key trait in ectotherms because it ties the rest of its biology together

Metabolism: sets metabolic rates and energy demand

Fecundity: larger females, more offspring

Life history: pace of growth and maturation

Mobility: Foraging range, speed

Research questions

Climatic vs. Habitat drivers:

Does body size track the temperature and rainfall as animal experiences or whether it lives above or underground?

Phylogenetic signal:

Is size structured by phylogenetic history or adaptation?

Tempo & Mode:

Is the stasis the pull of stabilizing selection towards one optimum or a generalist bauplan that succeeds everywhere?

Shape of the adaptive landscape

Testing alternative evolutionary models to body size

ACDC: *if body size evolved following an early burst*

Brownian Motion:
consistent with increase in variance through time

Brownian Motion with a Trend: *fossil informed (prior) analysis, consistente with Cope-Depéret's rule*

Single Peak vs. Multi-peak (habitat) OU: *body size evolved towards a single adaptive optimum*

Family Charontidae, sampled worldwide

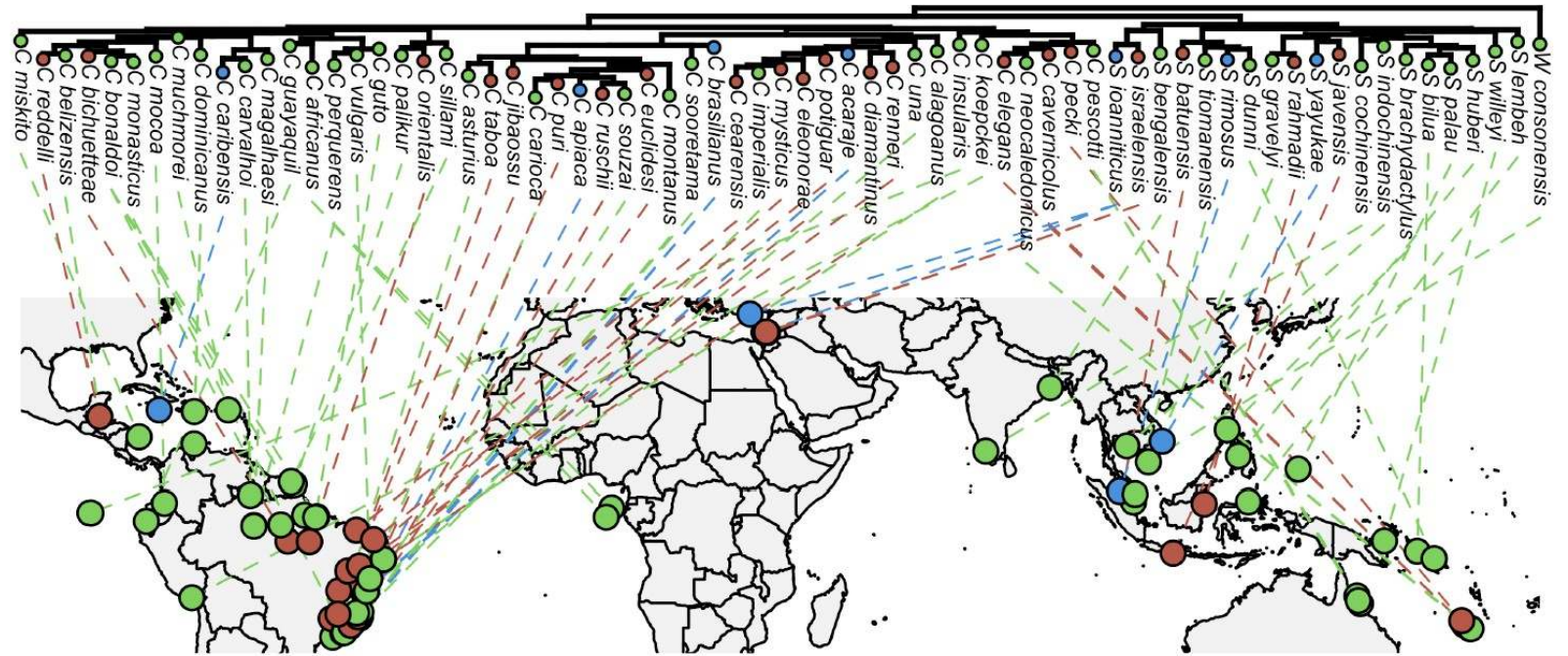
69

species · females only

Carapace length

body-size proxy

- Epigean – surface
- Troglophile – cave-associated
- Hypogean – strictly subterranean



- hypogean
- troglophile
- epigean

How we tested the drivers

INPUTS

Body size

log carapace length · 69 spp.

Habitat

epigean / troglophile / hypogean

Climate

bio1 · bio4 · bio12 · bio15

Phylogeny

dated, total-evidence tree



METHODS

Bayesian PGLMM

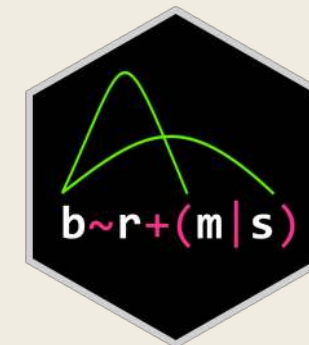
drivers, controlling for ancestry
and space

Blomberg's *K*

strength of phylogenetic signal

Model comparison

BM · Early Burst · OU



geiger
phytools

Syst. Biol. 73(3):495–505, 2024
© The Author(s) 2024. Published by Oxford University Press, on behalf of the Society of Systematic Biologists. All rights reserved. For commercial re-use, please contact reprints@oup.com
for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further
information please contact journals.permissions@oup.com.
<https://doi.org/10.1093/sysbio/syae021>
Advance Access Publication May 11, 2024

The Rediscovery of a Relict Unlocks the First Global Phylogeny of Whip Spiders (Amblypygi)

GUSTAVO S. DE MIRANDA¹, SIDDHARTH S. KULKARNI^{2,7}, JÉSSICA TAGLIATELA³, CAITLIN M. BAKER³,
ALESSANDRO P.L. GIUPPONI⁴, FACUNDO M. LABARQUE^{3,15}, EFRAT GAVISH-REGEV⁵, MICHAEL G. RIX⁶,
LEONARDO S. CARVALHO⁷, LÍVIA MARIA FUSARI⁸, MARK S. HARVEY^{9,10}, HANNAH M. WOOD¹, AND
PRASHANT P. SHARMA^{2,10}

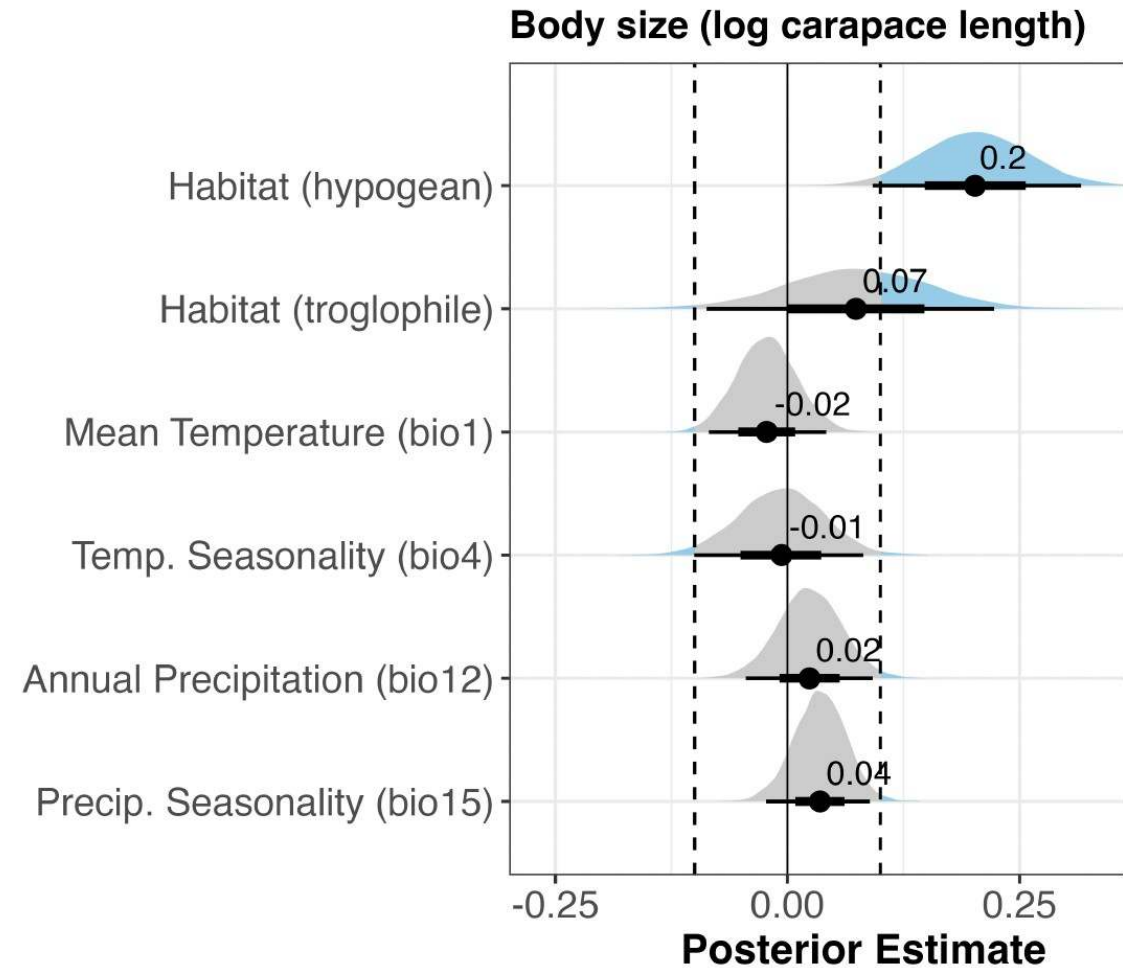
Habitat, not climate, predicts body size

Only **hypogean habitat** falls outside the region of practical equivalence. **22% LARGER IN HYPOGEAN**

Every climate term — temperature, seasonality, precipitation — stays inside the ROPE. **0.2% SIZE CHANGE PER 1°C (BIO1)**

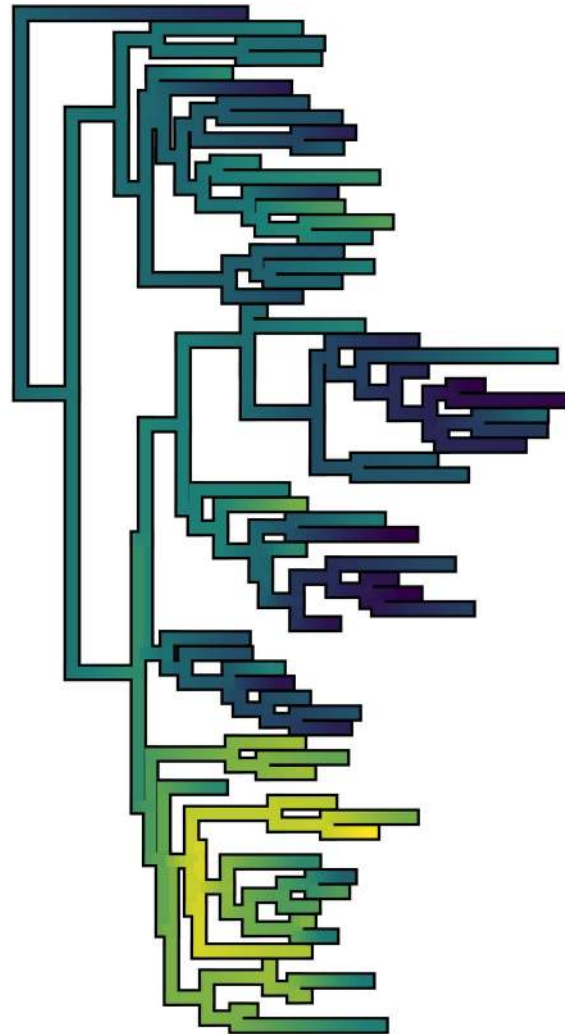
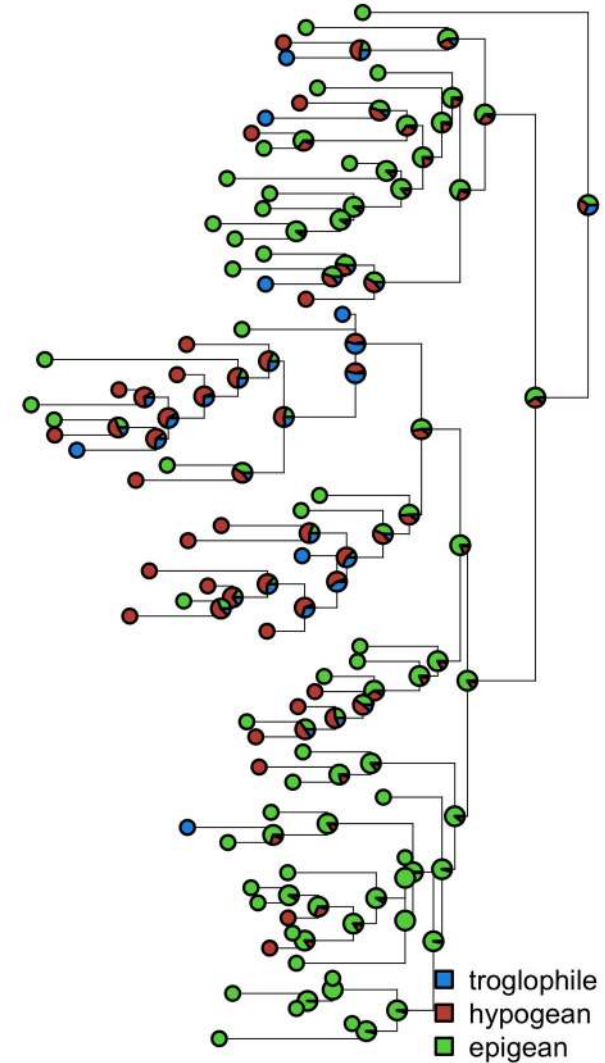
Posterior · log carapace length fixed effects explain $\approx 21\%$ of variance

Species occupy thermally buffered microhabitats (fissures/rocks), decoupling them from regional macroclimatic gradients



Blomberg's K · moderate phylogenetic signal

Going underground brings no acceleration of size evolution.

[illegible]

RESULT · MODE

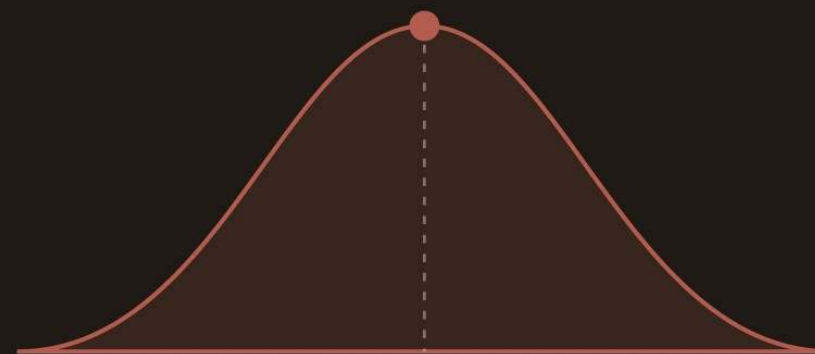
One adaptive peak, not many

$w_i =$	$w_i =$	$w_i =$
0.006	0.002	0.949

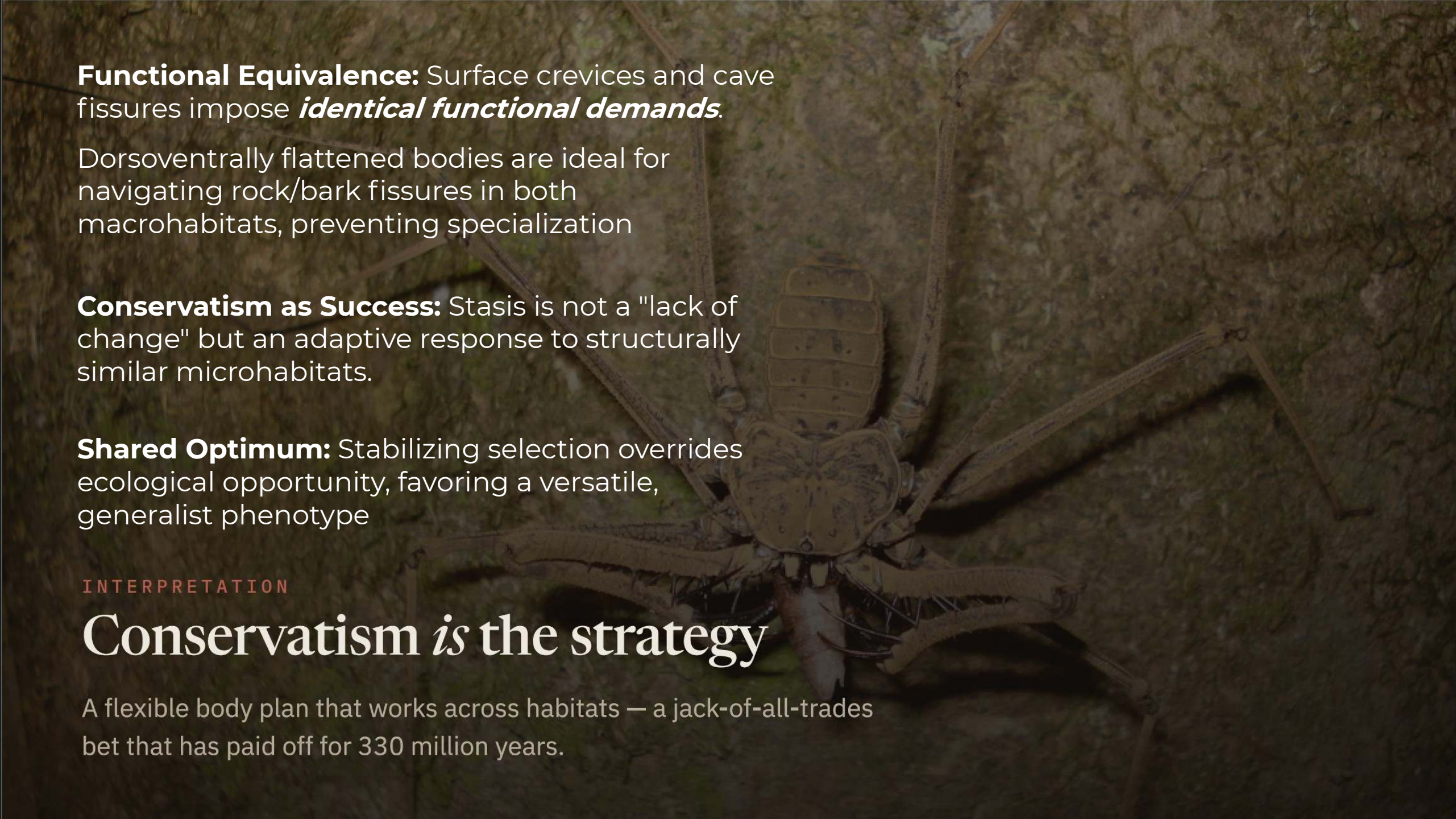


Body size is pulled toward **one shared optimum** across the whole family — stabilizing selection in deep time.

Morphological conservatism is maintained by stabilizing selection *across habitats*.



adaptive landscape · one peak



Functional Equivalence: Surface crevices and cave fissures impose *identical functional demands*.

Dorsoventrally flattened bodies are ideal for navigating rock/bark fissures in both macrohabitats, preventing specialization

Conservatism as Success: Stasis is not a "lack of change" but an adaptive response to structurally similar microhabitats.

Shared Optimum: Stabilizing selection overrides ecological opportunity, favoring a versatile, generalist phenotype

INTERPRETATION

Conservatism *is* the strategy

A flexible body plan that works across habitats — a jack-of-all-trades bet that has paid off for 330 million years.

Acknowledgements

QR Code for 10.1101/2025.10.06.680801v1

The impact of climate and habitat on body shape and size evolution in whip spiders (Amblypygi)



Received minor
revisions in



Sam Noble Museum



FUNDAÇÃO
UNIVERSIDADE
FEDERAL DE
MATO GROSSO DO SUL



Fundação de Apoio ao Desenvolvimento do Ensino,
Ciência e Tecnologia do Estado de Mato Grosso do Sul

